

The logo for Oracle Academy is centered on a light gray background. It features the word "ORACLE" in a bold, orange, sans-serif font. Below it, the word "Academy" is written in a smaller, dark gray, sans-serif font. The entire logo is framed by a thin black border, with dark gray horizontal bars at the top and bottom.

ORACLE

Academy

Database Foundations

3-3

Normalization and Business Rules

ORACLE
Academy



Copyright © 2020, Oracle and/or its affiliates. All rights reserved.

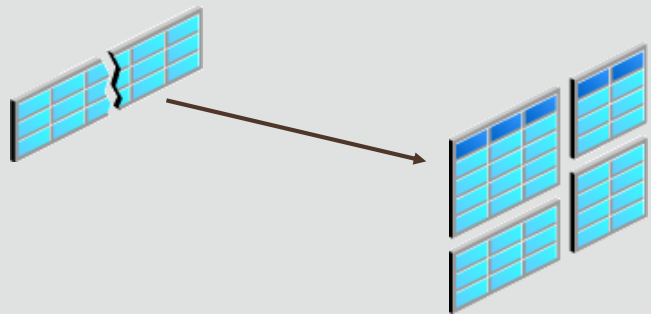
Objectives

- This lesson covers the following objectives:
 - Explain normalization
 - Describe the normal forms
 - Use normalization to validate data
 - Describe business rules



Normalization

- Is the process of organizing the attributes and tables of a relational database to minimize redundancy
- Helps in handling insert, update, and delete anomalies, ensuring a better performance of the database



Normalization can be defined as the process of decomposing a table with inconsistencies to produce a smaller, well-structured table. Normalization may involve splitting a table into two or more tables and defining relationships between the tables. The objective is to isolate data so that additions, deletions, and modifications of an attribute or a field can be made in just one table and then propagated throughout the rest of the database by using well-defined relationships. Edgar F. Codd, the inventor of the relational model, introduced the concept of normalization.

Why Should You Normalize Data?

- Reduce redundant data in the existing design
- Increase the integrity of data and the design's stability
- Eliminate other types of data inconsistencies and anomalies
- Identify missing tables, columns, and constraints



No Duplicate Content!



ORACLE
Academy

DFO 3-3
Normalization and Business Rules

Copyright © 2020, Oracle and/or its affiliates. All rights reserved.

5

What Is Normalization?

- Normalization is a relational database concept, but its principles apply to data modeling
- The goal is to normalize data to 3NF before transforming the model to your relational design

Rule	Description
First Normal Form (1NF)	All attributes must be single-valued.
Second Normal Form (2NF)	An attribute must be dependent on its entity's entire UID.
Third Normal Form (3NF)	No non-UID attributes can be dependent on another non-UID attribute.

Project Exercise 1

- DFo_3_3_1_Project
 - Oracle Baseball League Store Database
 - Converting the Data into Unnormalized Form



Types of Normal Forms

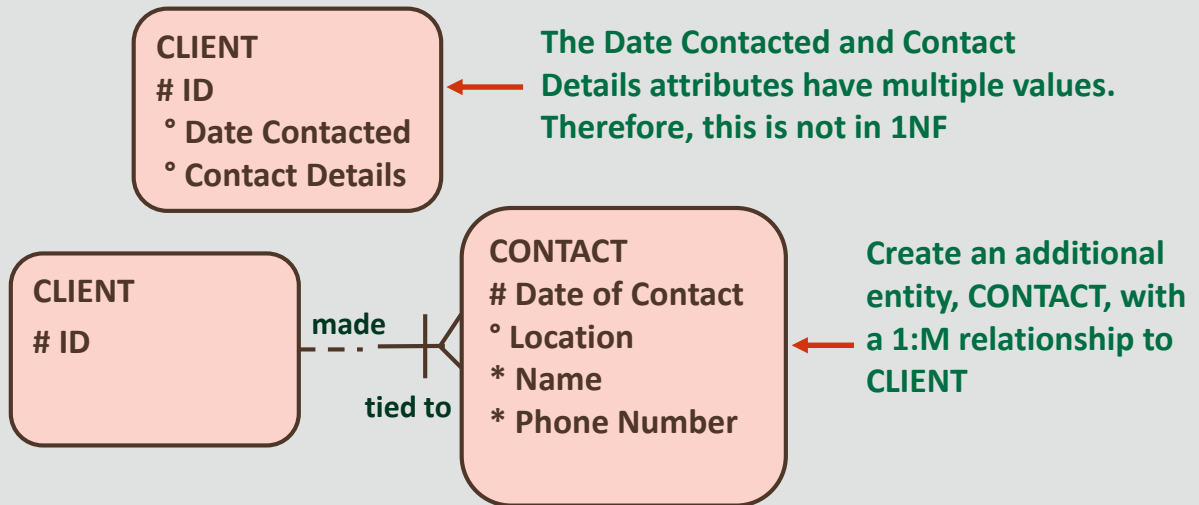
- Main types of normal forms introduced by E.F. Codd:
 - First Normal Form (1NF)
 - Second Normal Form (2NF)
 - Third Normal Form (3NF)

What is First Normal Form? (1NF)

- First Normal Form requires that no multi-valued attributes exist
- To check for 1NF, validate that each attribute has a single value for each instance of the entity
- If an attribute is multi-valued, create an additional entity and relate it to the original entity with a 1:M relationship

First Normal Form (1NF)

- Each attribute must have a single value for each occurrence of the entity



Project Exercise 2

- DFo_3_3_2_Project
 - Oracle Baseball League Store Database
 - First Normal Form

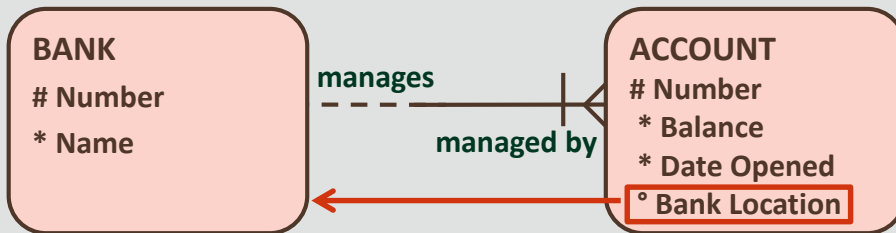


What is Second Normal Form? (2NF)

- Second Normal Form (2NF) requires that any non-UID attribute be dependent on (be a property of, or a characteristic of) the entire UID
- If the UID is a composite UID each attribute must be dependent on all parts of the composite UID
- If an attribute is not dependent on the entire UID, create an additional entity with the partial UID

Second Normal Form (2NF)

- An attribute must be dependent on its entity's entire UID



**The Bank Location attribute is dependent on BANK , not on ACCOUNT.
Therefore, this is not in 2NF. Move the attribute to the BANK entity.**

Project Exercise 3

- DFo_3_3_3_Project
 - Oracle Baseball League Store Database
 - Second Normal Form

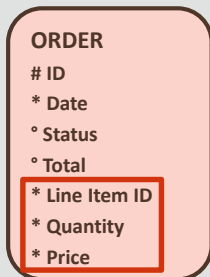


What is Third Normal Form? (3NF)

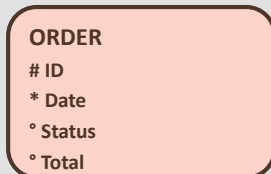
- The rule of Third Normal Form (3NF) states that no non-UID attribute can be dependent on another non-UID attribute
- Third Normal Form prohibits transitive dependencies
- A transitive dependency exists when any attribute in an entity is dependent on any other non-UID attribute in that entity
- You need to move any non-UID attribute that is dependent on another non-UID attribute into a new entity

Third Normal Form (3NF)

- Each attribute depends only on the UID of its entity
- Move any non-UID attribute that is dependent on another non-UID attribute into a new entity

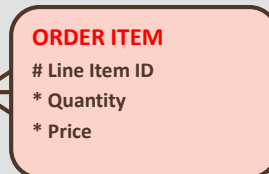


The Quantity and Price attributes are dependent on the ORDER table ID (UID) and Line Item ID (non-UID). Therefore, this is not in 3NF.



contains

included in



Create a new ORDER ITEM entity. Move the Line Item ID, Quantity, and Price attributes to the new entity, and then create an identifying relationship.

Project Exercise 4

- DFo_3_3_4_Project
 - Oracle Baseball League Store Database
 - Third Normal Form



Business Rules

- A business rule is a statement that defines or constrains some aspect of the business
- It is used to define entities, attributes, relationships, and constraints
- It includes rules and policies that define how a business operates and handles its day-to-operations

Business Rules

- There are two types of business rules:
 - Structural
 - Procedural
- Example: All orders must have a Contact Number

The example in the slide is a structural business rule.

Sources of Business Rules

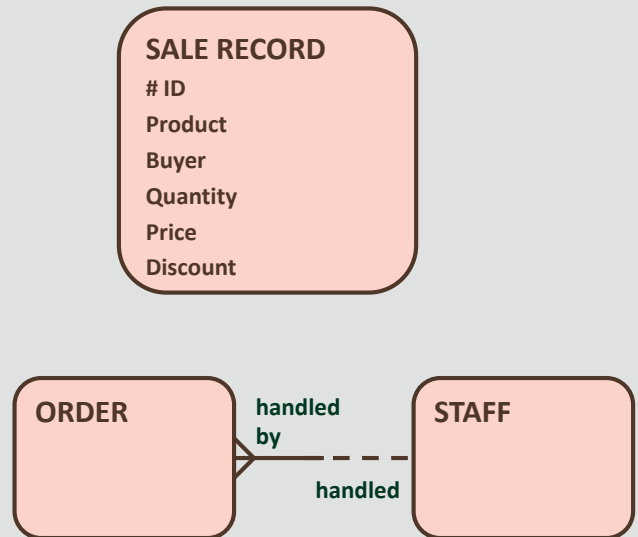
- Top management and managers
- Written documentation
 - Procedures
 - Standards
 - Operations manuals
- Direct interviews with end users

Structural Business Rules

- Structural business rules indicate the types of information to be stored and how the information elements interrelate
- Structural business rules detail a specific, static aspect of the business
- These rules can always be diagrammed in the ERD

Structural Business Rules: Examples

- A sales record includes product, buyer, quantity, price, and discount
- All restaurant orders must be handled by a staff member (specifically, an order taker)



Library Case Scenario – Structural Rules

Is it
reasonable/effective
for two members to
have the same
member ID or no
member ID?

Is it
reasonable/effective
for two books to have
the same book ID
number or no book ID
number at all?



Procedural Business Rules

- Procedural rules deal with the prerequisites, steps, processes, or workflow requirements of a business
- Many procedural business rules are related to time: Event A must happen before event B
- Some procedural business rules cannot be diagrammed, but must still be documented so that they can be programmed later

Procedural Business Rules: Examples

- An online store might not accept a next-day delivery order if the order is received after 3:00 PM
- Approval for travel requests to a conference must be signed by the manager

Library Case Scenario – Procedural Rules

**Books must be
returned to the
library within
30 days**

**Members who do
not return the book
by the due date will
not be permitted to
borrow another
book**



Additional Programming

- Not all business rules can be modeled when a conceptual data model is being developed. Some rules, such as these must be implemented by programming as processes that interact with data:
 - Any employee whose overtime exceeds 10 hours per week must be paid 1.5 times the hourly rate
 - OR
 - Customers whose account balances are 90 days overdue will not be permitted to charge additional orders

Summary

- In this lesson, you should have learned how to:
 - Explain normalization
 - Describe the normal forms
 - Use normalization to validate data
 - Describe business rules



The logo for Oracle Academy is centered on a light gray background. It features the word "ORACLE" in a bold, orange, sans-serif font. Below it, the word "Academy" is written in a smaller, dark gray, sans-serif font. The entire logo is framed by a thin black border, with dark gray horizontal bars at the top and bottom.

ORACLE

Academy